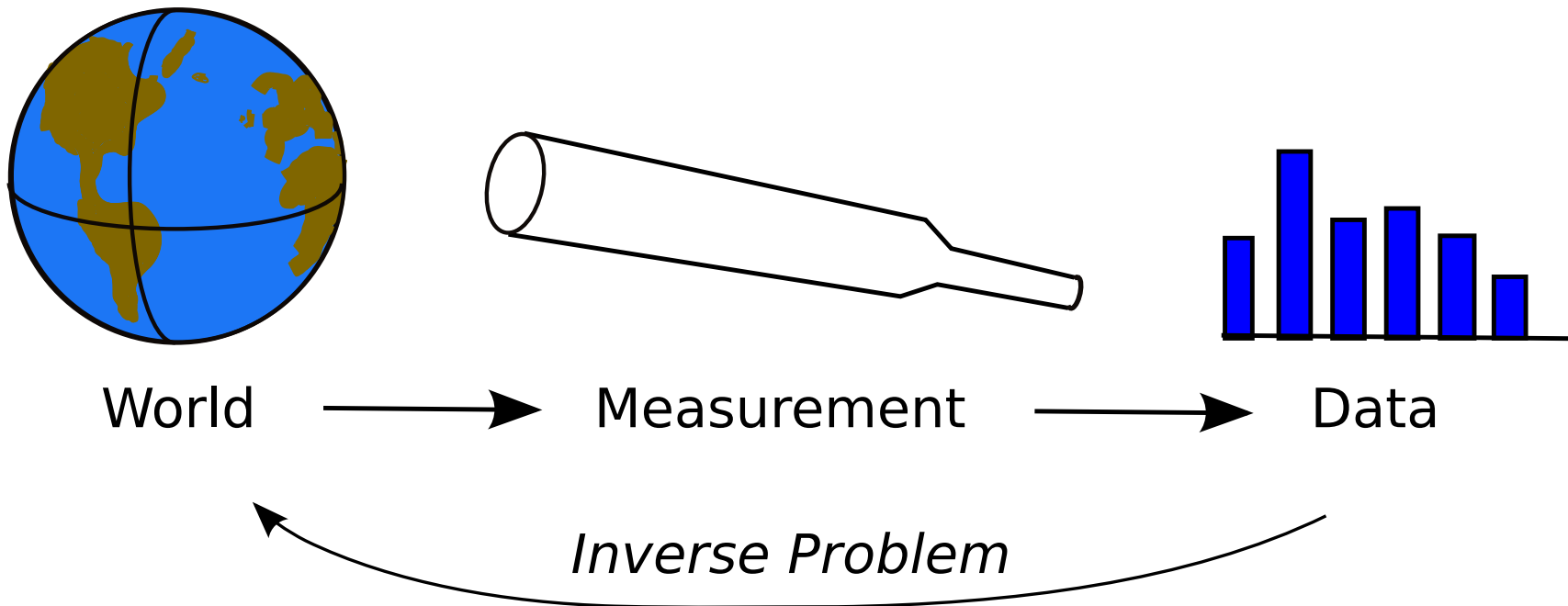


Mathematics of Machine Learning

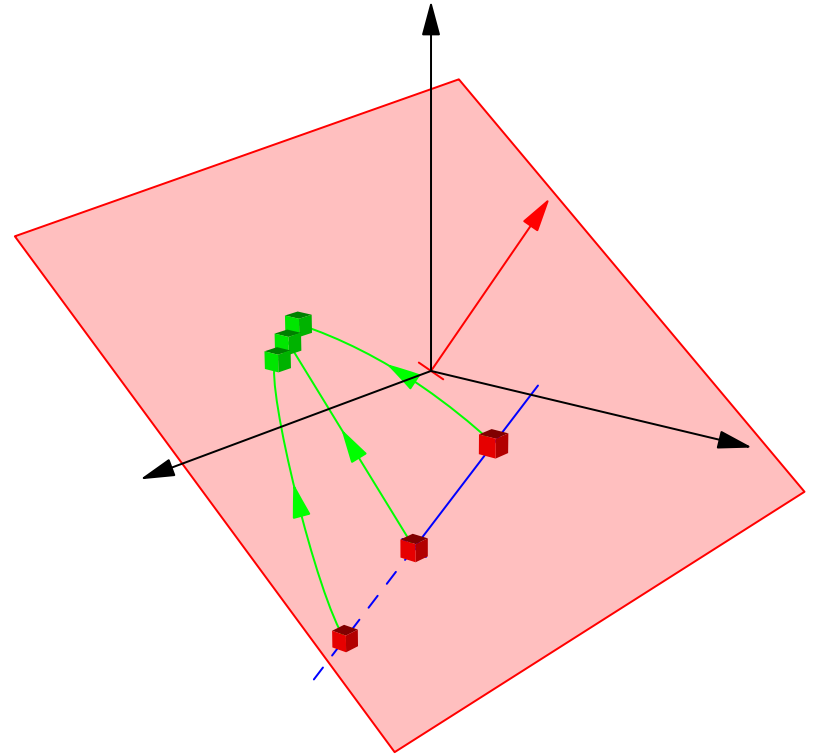
Understand Mappings



Mappings, Linear Maps, Solving Linear Systems

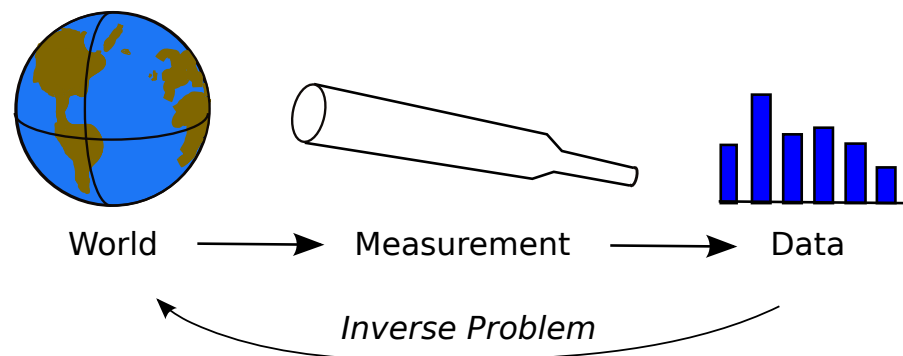
Outline

1. **Mappings**
2. Linear Maps



Transforming Data

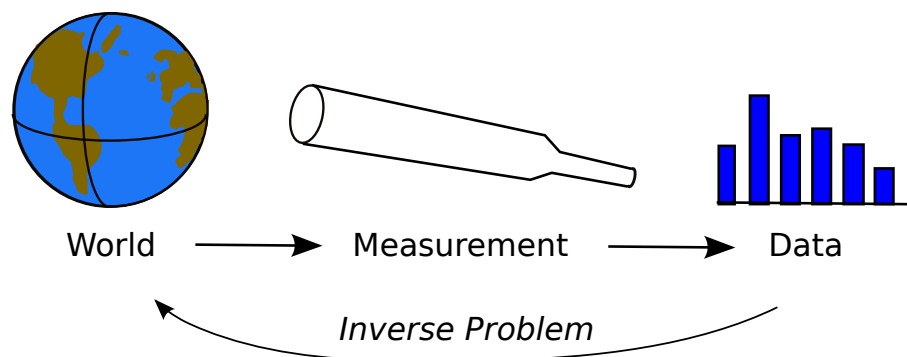
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- At a mathematical level machine learning can be viewed as performing an inverse mapping



- Although our mappings are not necessarily linear in either direction we learn a lot by understanding linear operators

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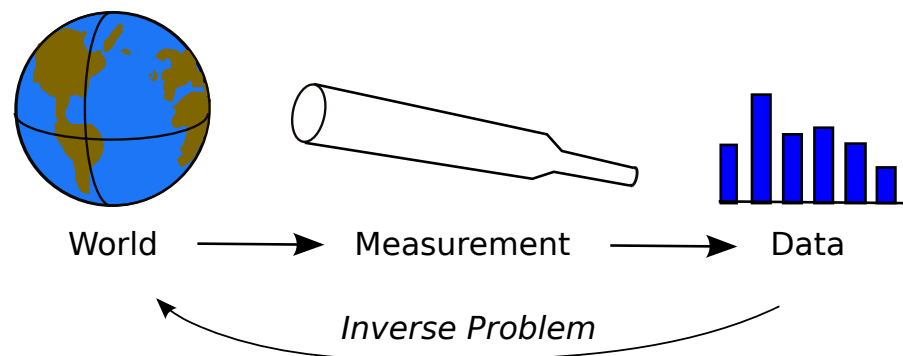
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Inverse Problems

- Given m observations $\{(\mathbf{x}_k, y_k) | k = 1, \dots, m\}$ and p unknown $\mathbf{w} = (w_1, w_2, \dots, w_p)^\top$ such that $\mathbf{x}_k^\top \mathbf{w} = y_k$ then to find $\mathbf{w}$
- Define the *design matrix* as the matrix of feature vectors

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mp} \end{pmatrix}$$

- and the target vector $\mathbf{y} = (y_1, y_2, \dots, y_m)^\top$
- Then if $m = p$ we have $\mathbf{y} = \mathbf{X}\mathbf{w}$ or $\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$

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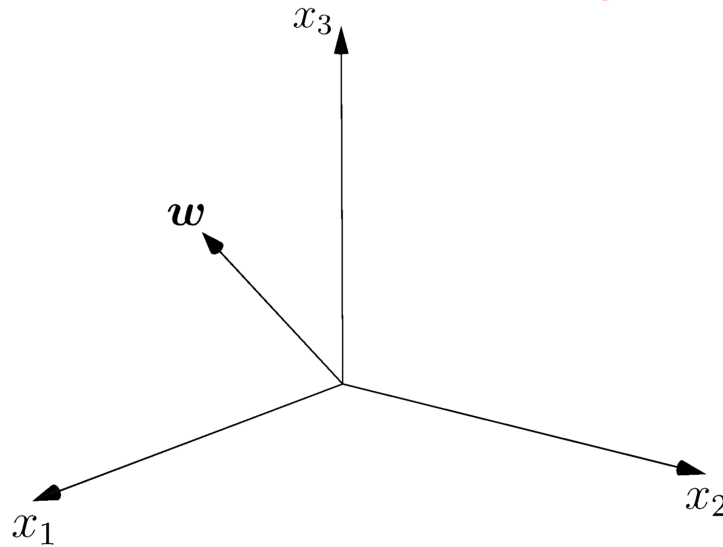
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Linear Regression

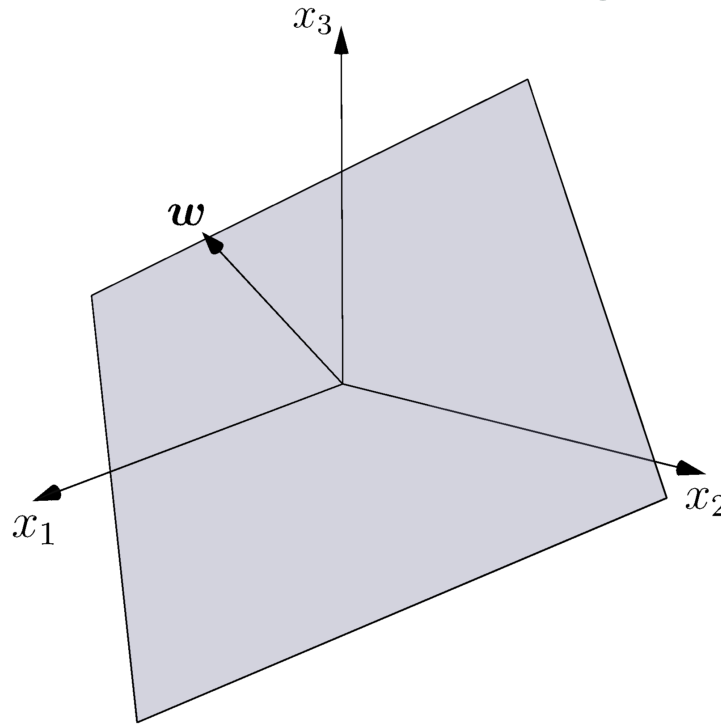
- $x_k^T w$ depends on distance from separating



- If $m > p$ then $\mathbf{X}$ isn't square so doesn't have an inverse
- Worse unless the data is accurate $y \approx \mathbf{X}w \Rightarrow$ no “solution”
- Problem solved by Gauss to predict the orbit of the asteroid Ceres

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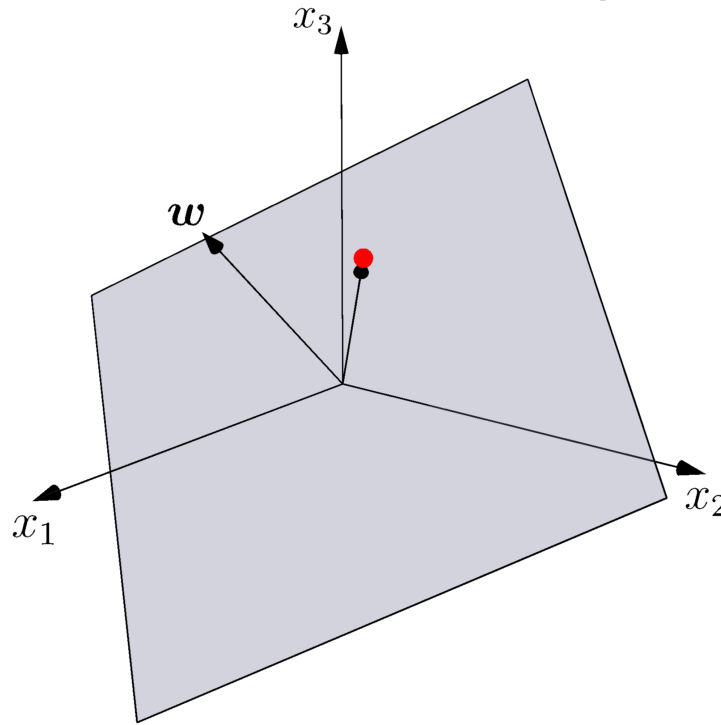
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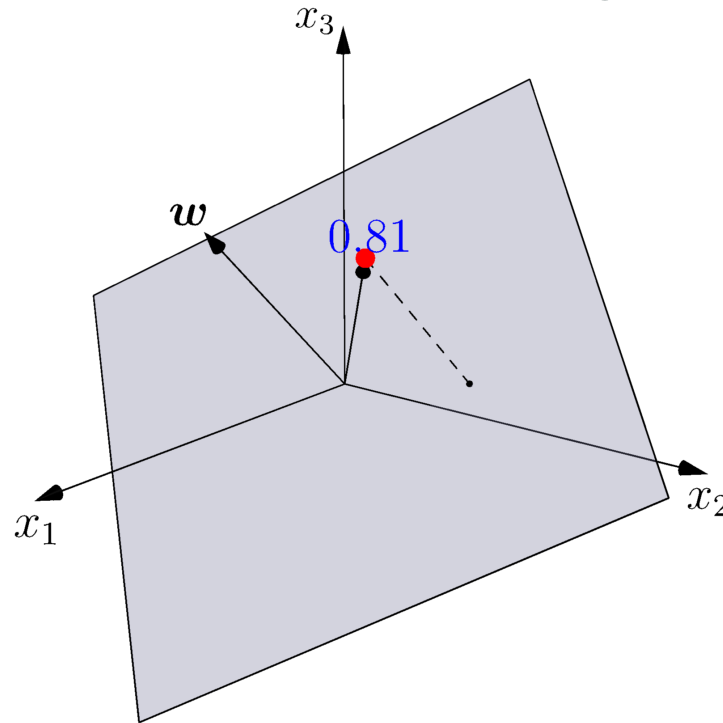
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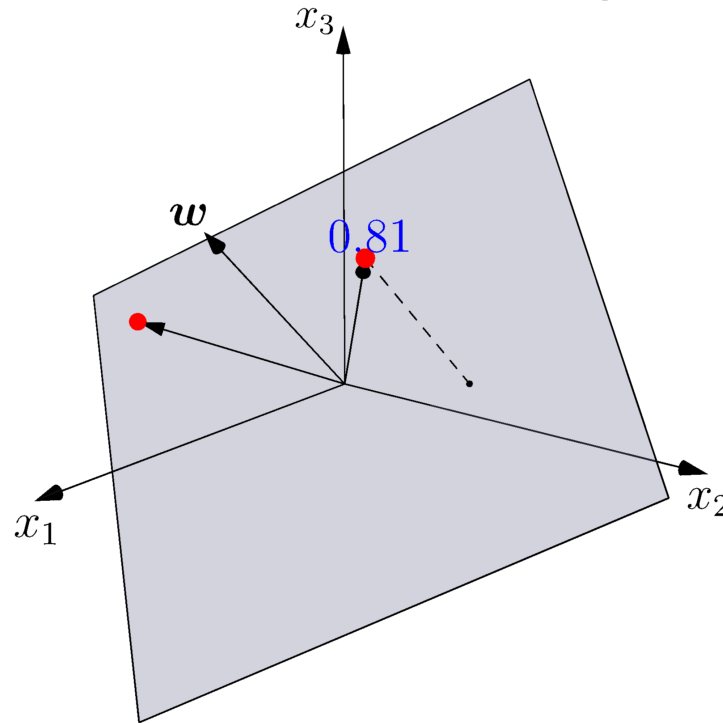
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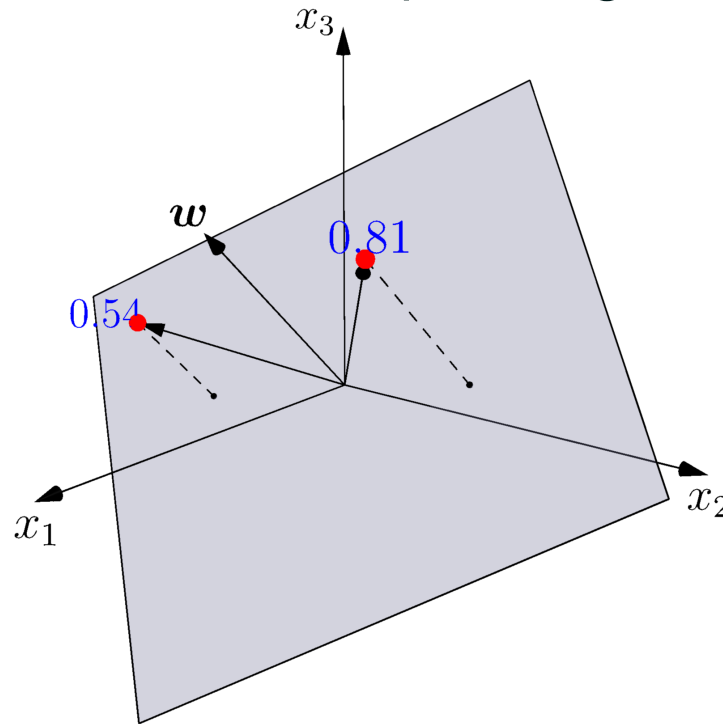
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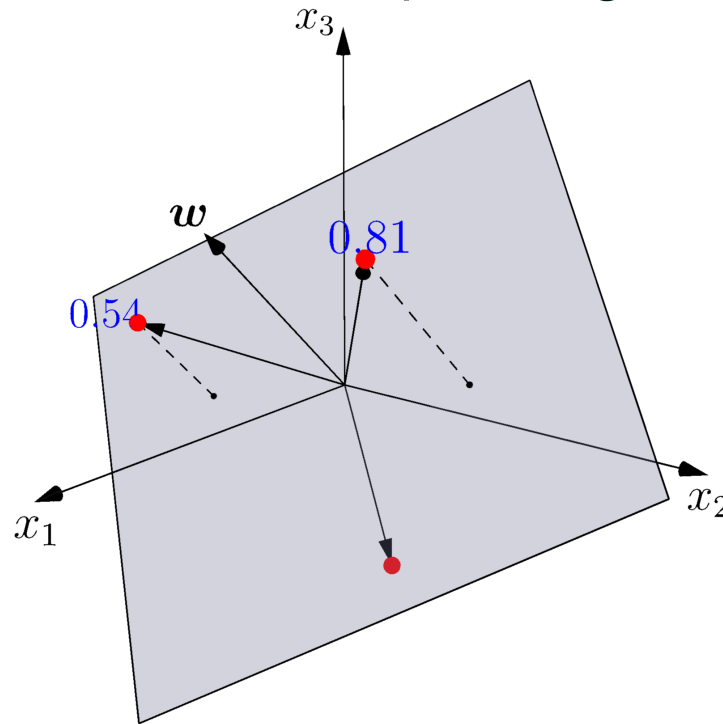
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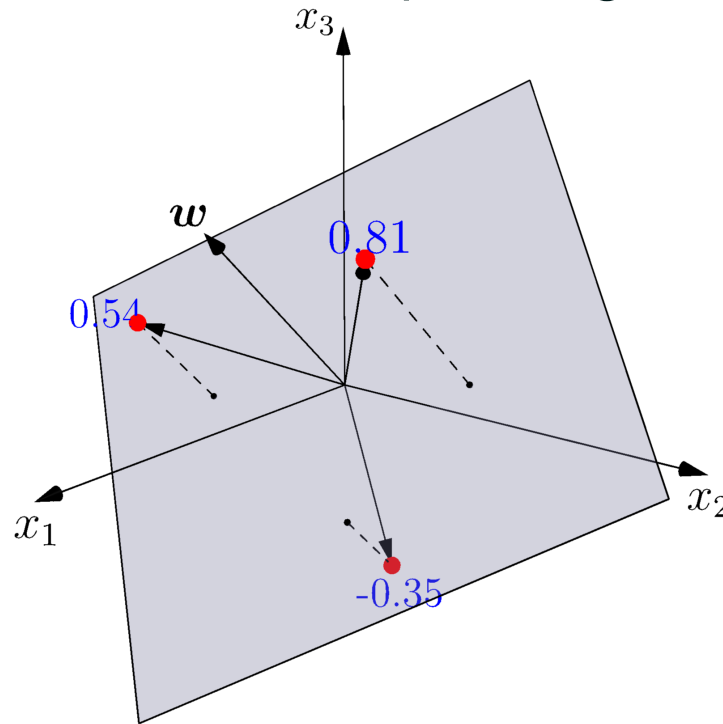
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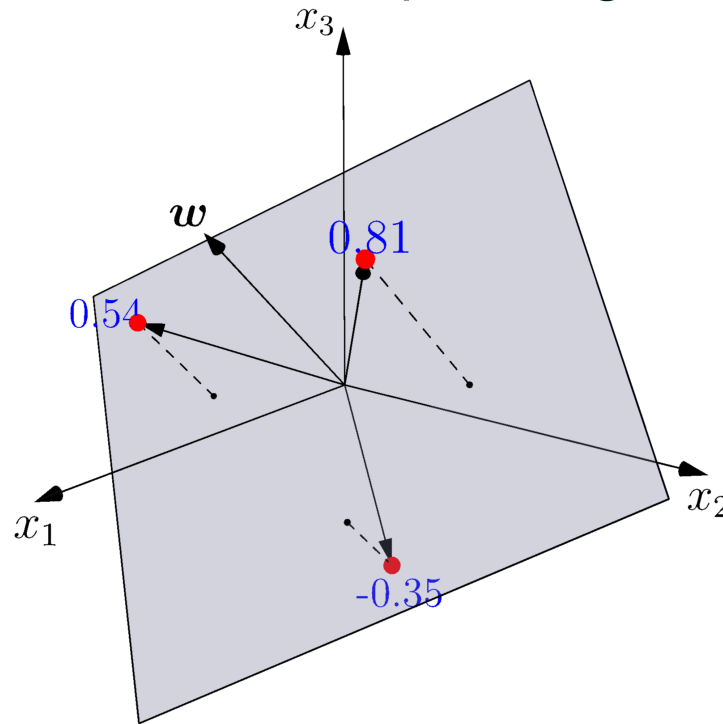
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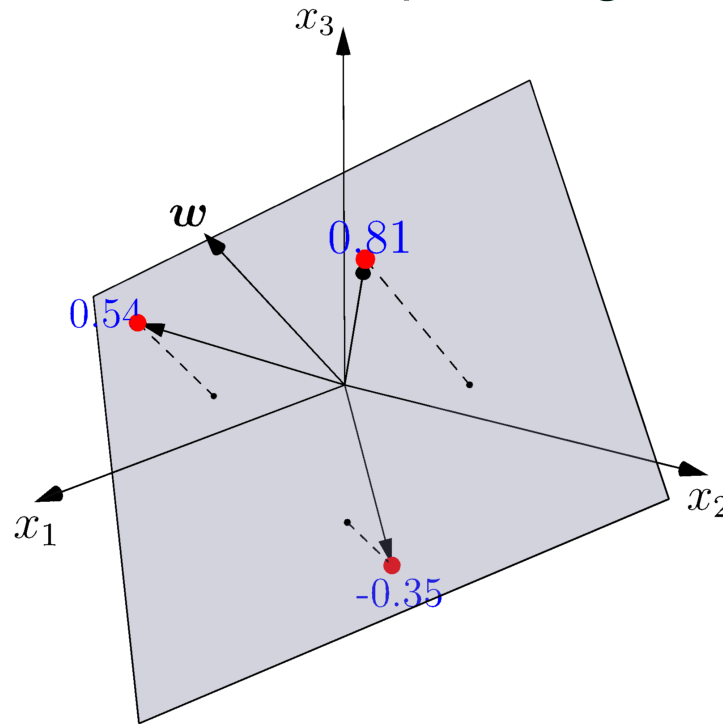
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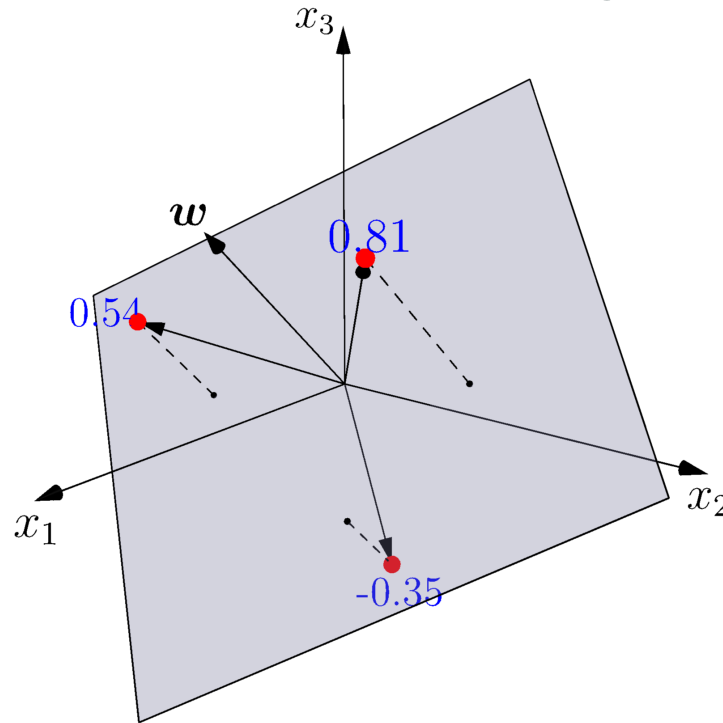
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Linear Least Squares

- The error of input pattern $\mathbf{x}_k$ is

$$\epsilon_k = \mathbf{x}_k^\top \mathbf{w} - y_k$$

- The squared error

$$E(\mathbf{w}|\mathcal{D}) = \sum_{k=1}^m (\mathbf{x}_k^\top \mathbf{w} - y_k)^2 = \sum_{k=1}^m \epsilon_k^2 = \|\boldsymbol{\epsilon}\|^2$$

- We can define the error vector

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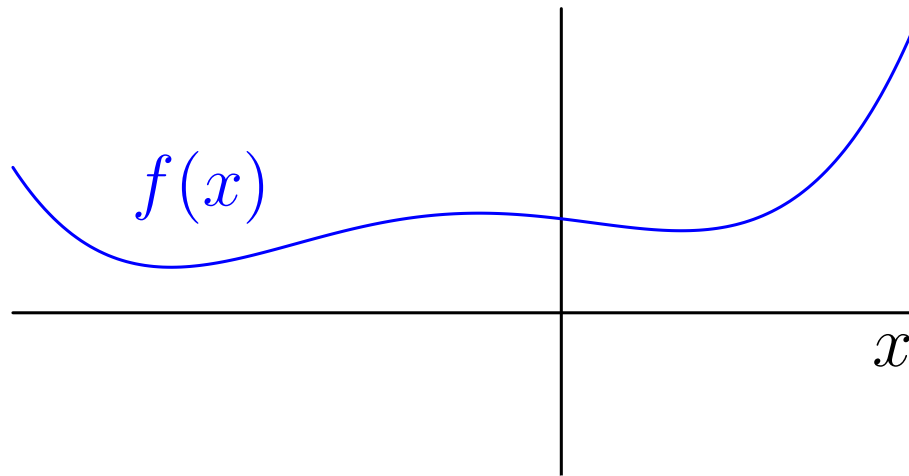
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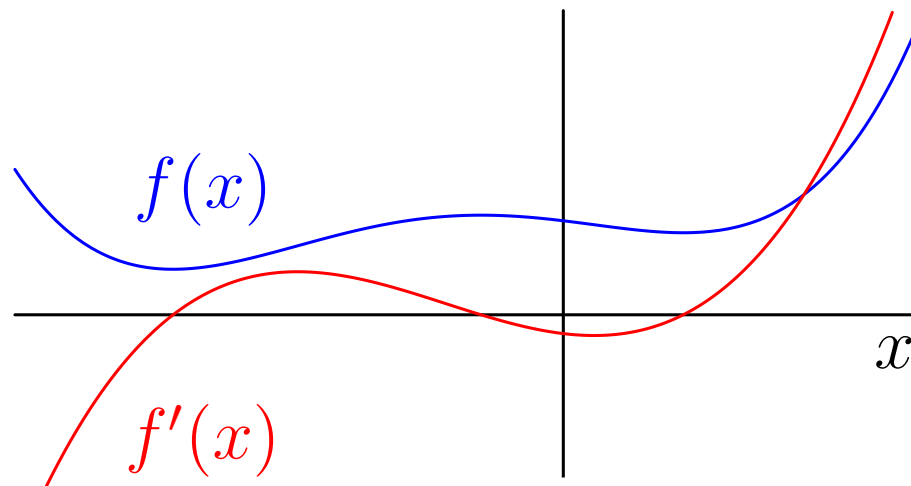
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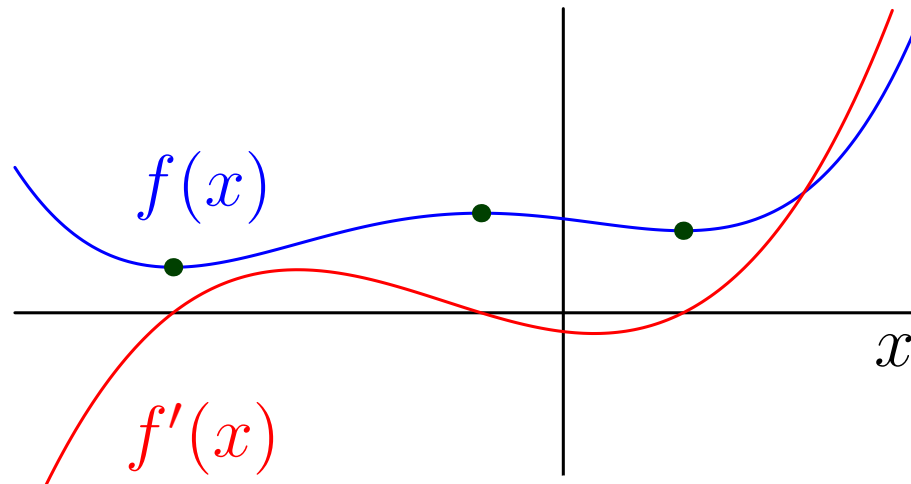
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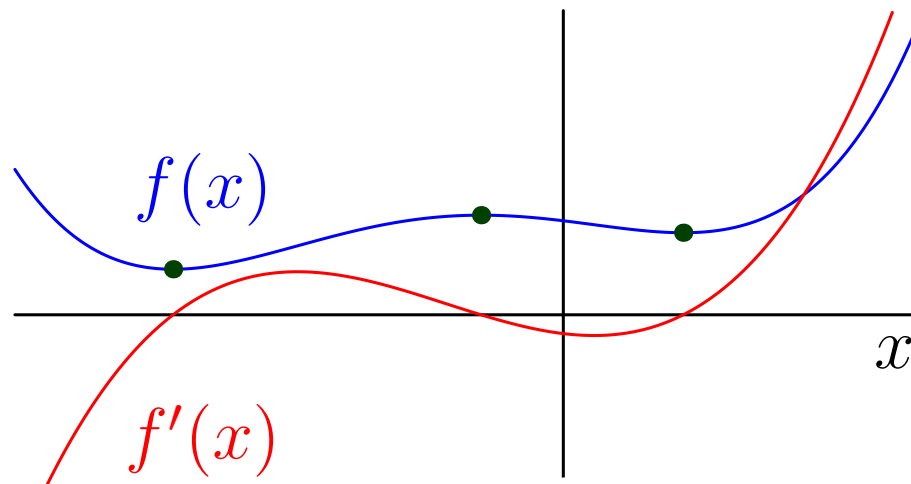
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- The minima of an n -dimensions function $f(\mathbf{x})$ are given by the set of equations

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0 \quad \forall i = 1, \dots, n$$

Gradients

- The **grad** operator ∇ is the gradient operator in high dimensions

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

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Least Squares Solution

- The least squared solution is give by

$$\begin{aligned}\nabla E(\boldsymbol{w}|\mathcal{D}) &= \nabla \|\boldsymbol{\epsilon}\|^2 = \nabla \|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2 \\ &= \nabla (\boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} - 2\boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}) \\ &= 2(\boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} - \boldsymbol{X}^\top \boldsymbol{y}) = 0\end{aligned}$$

- Or

$$\boldsymbol{w} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{X}^+ \boldsymbol{y}$$

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Missing Bits of the Mathematics

- Note that $\|\mathbf{a}\|^2 = \mathbf{a}^\top \mathbf{a} = \sum_i a_i^2$

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Computing Gradients

- To understand gradients we sometimes need to go back to components

$$\nabla w^T M w$$

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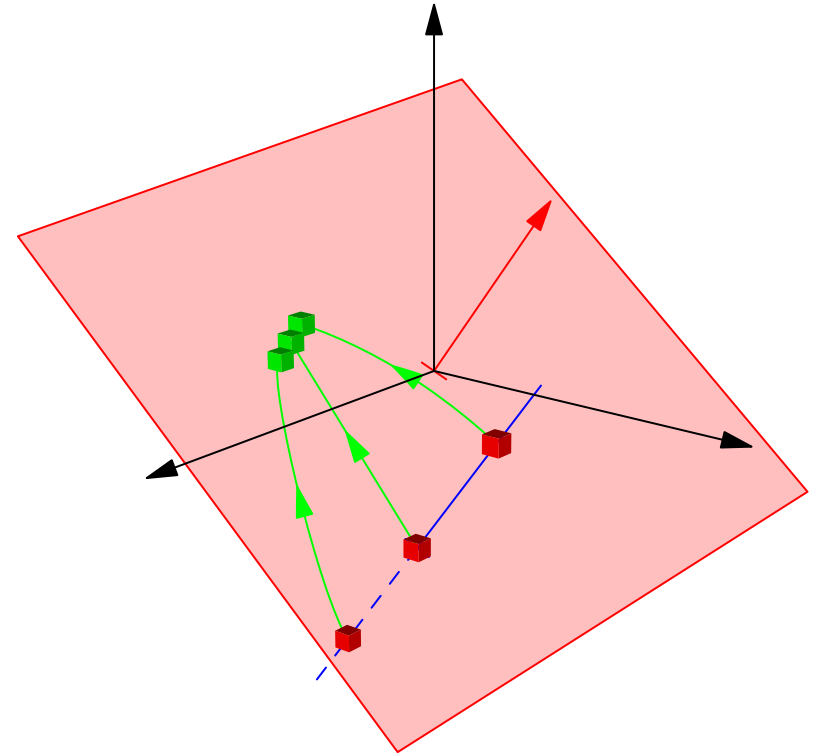
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- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

Outline

1. Mappings
2. **Linear Maps**



Solving Inverse Problems

- Gauss showed us how to solve **over-constrained** problems (we have more observations than parameters)
- We seek a solution which isn't necessarily exact but minimises an error
- But, what if we have more parameters than observations
- That is, we are **under-constrained**
- Note that in some directions you might be over-constrained and in other directions under-constrained

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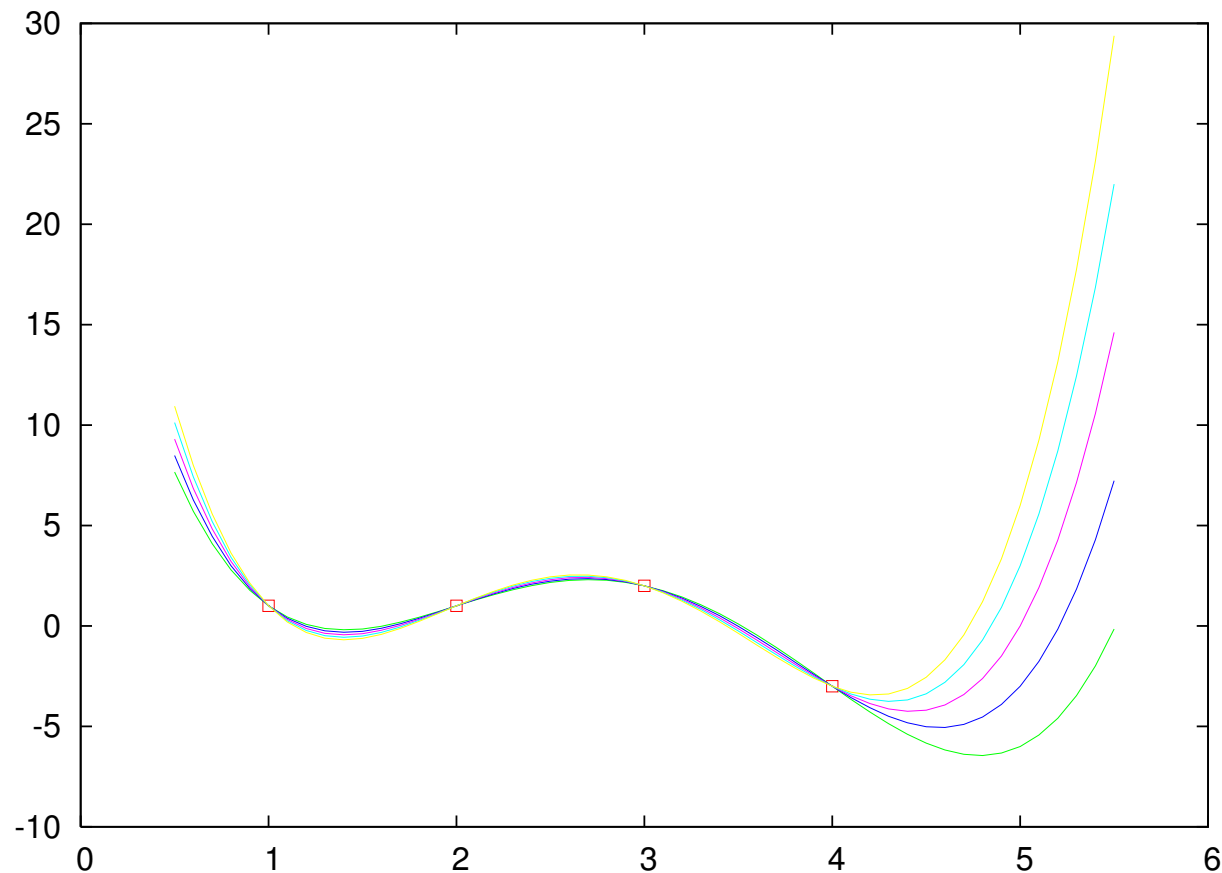
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- This is very typical of most machine learning problems

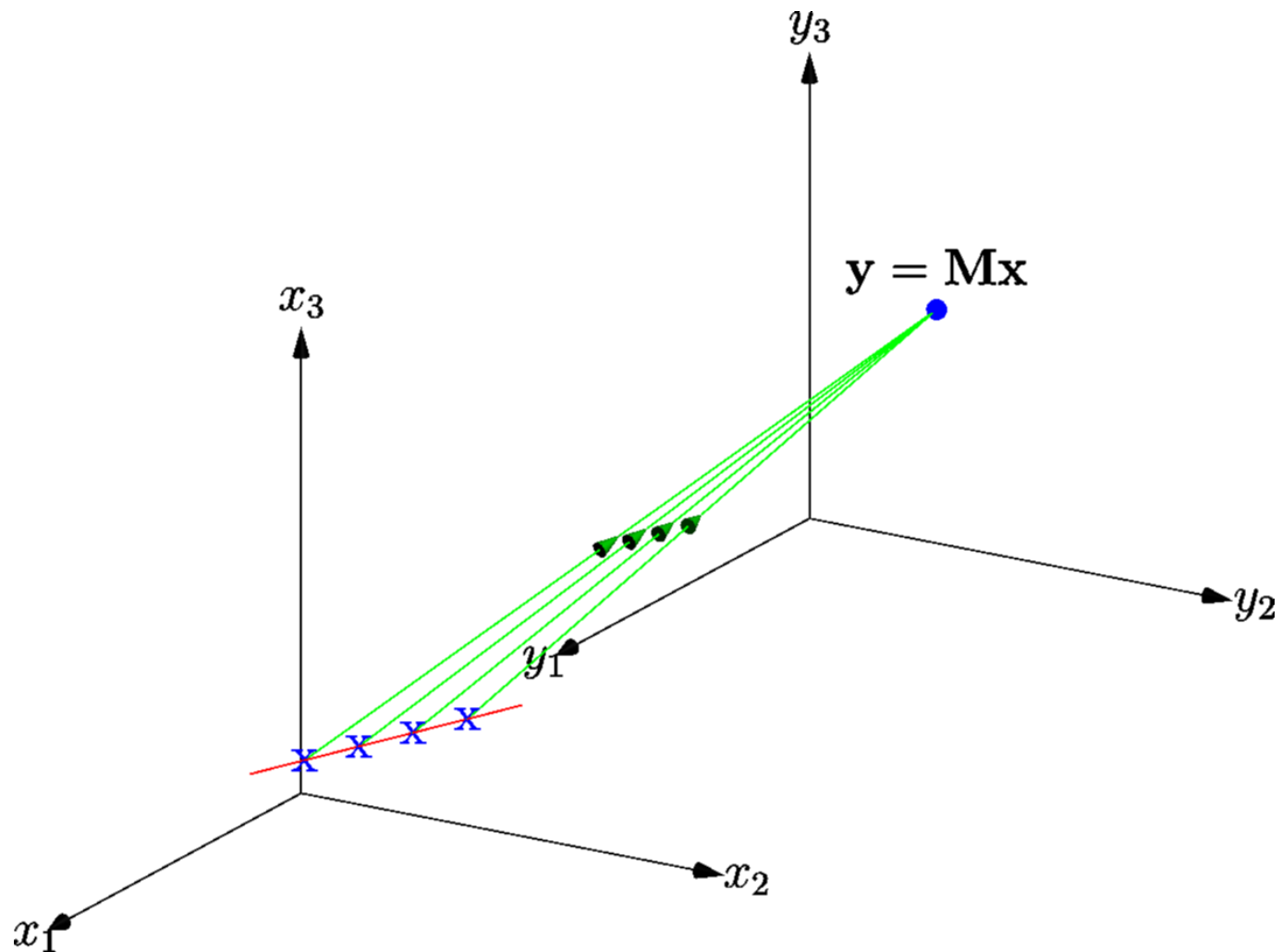
Under Constrained Systems

- If we have less data-points than parameters then there will be multiple solutions



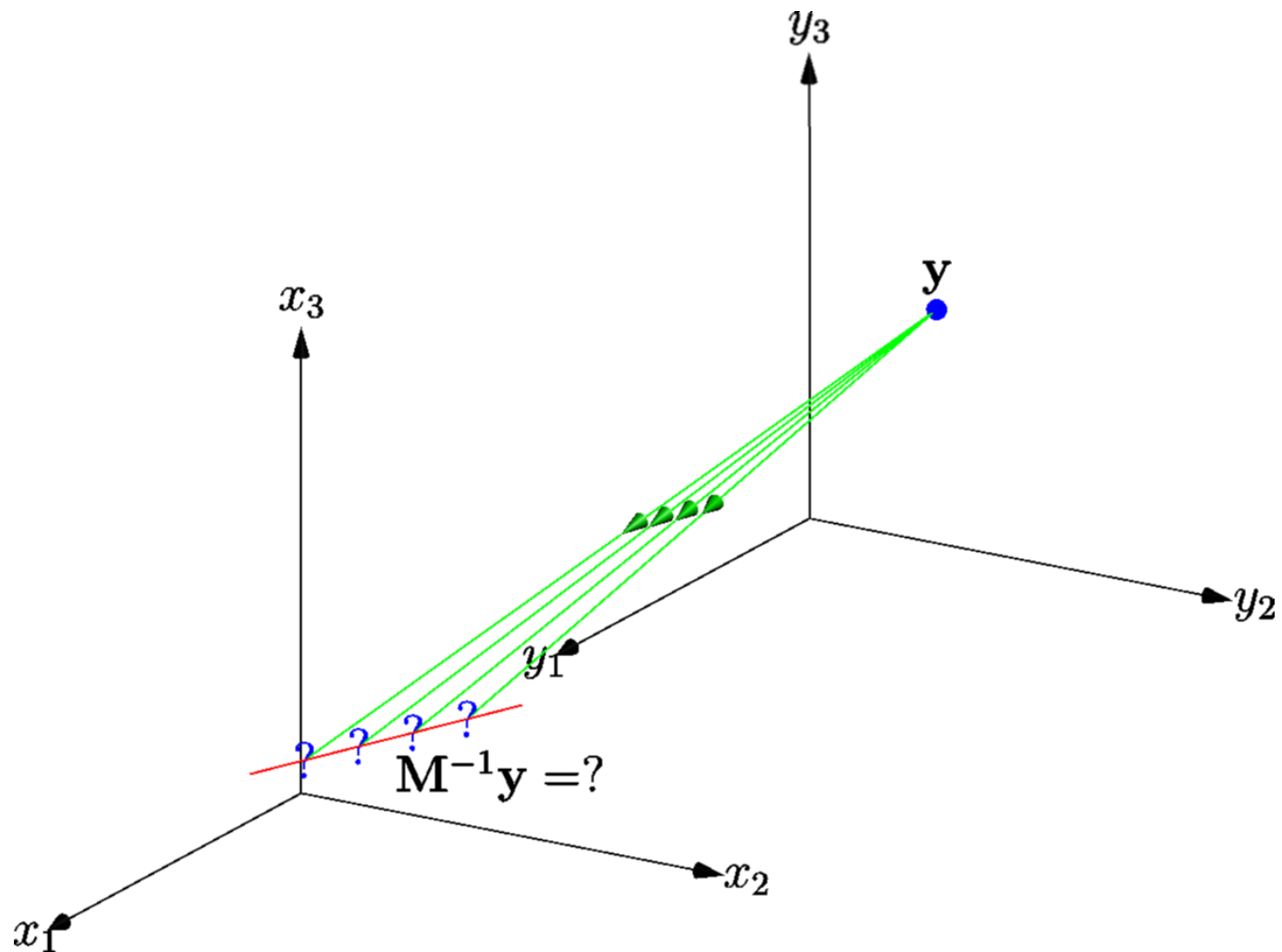
What is the Inverse?

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- The system is **under-constrained**
- We have more unknowns than equations
- The inverse is not unique
- Solving the inverse problem ($w = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$) is said to be **ill-posed**
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III-Conditions

- Singular matrices are rare (although they occur when we don't have enough data), but matrices that are close to being singular are common
- If a matrix is close to singular it is ill-conditioned
- Ill-conditioned matrices have some small eigenvalues
- All points get contracted towards a plane
- Large matrices are very often ill conditioned

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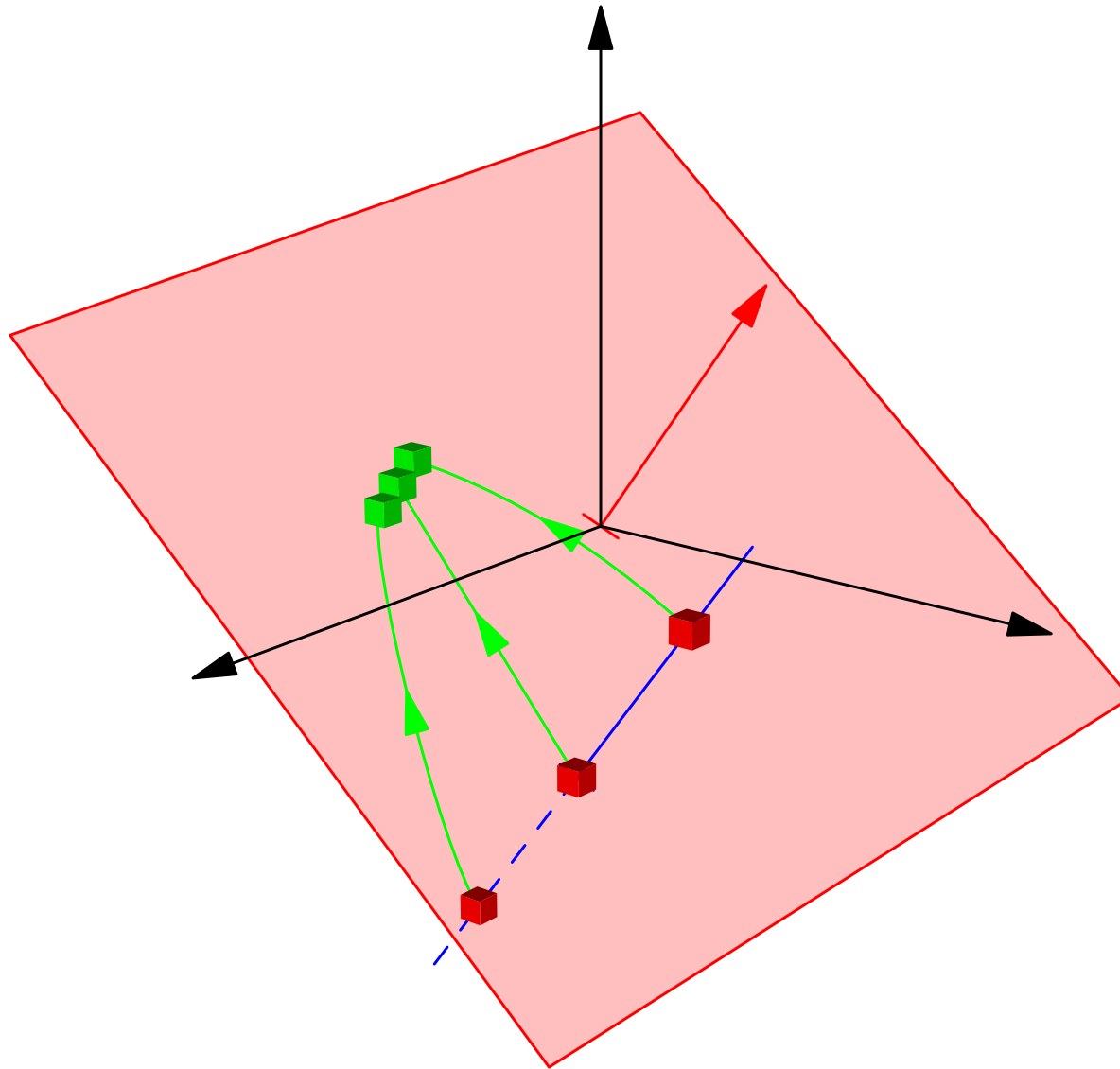
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III-Conditioning in ML

- Ill-conditioning in machine learning occurs when a very small change in the learning data causes a large change in the predictions of the learning machine
- In linear regression the matrix $\mathbf{X}^T\mathbf{X}$ is typically ill-conditioned when we have as many data points as parameters
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